

# Devismita Panda

Doctoral Student, Laboratoire d'Astrophysique De Bordeaux, University of Bordeaux, France

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## EDUCATION

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### PhD, Infinity<sup>2</sup> International Graduate Program

Sept 2024 – Ongoing

*Laboratoire d'Astrophysique De Bordeaux (LAB), Bordeaux, France*

**Thesis:** "Origin of Complex Organic Molecules (COMs) in Massive Star-Forming Regions" (Data: NASCENT-stars large program)

**Supervisor:** Dr. Timea Csengeri

### M.Sc. Physics (Specialization: Astronomy and Astrophysics)

Oct 2020 – Jun 2022

*Pondicherry University, Puducherry, India*

**Result:** 90.07% with Distinction

### B.Sc. Honors in Physics

Aug 2016 – Jun 2019

*Government Science College, Chatrapur, Odisha, India*

**Result:** 80.08% with Distinction

## RESEARCH EXPERIENCE

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### Visiting Researcher

Mar 2026 – May 2026

*I. Institute of Physics (PH1), University of Cologne, Germany*

**Project:** "Spectral Modeling of Complex Organic Molecules in CygX-N30 with XCLASS" (Funded by DAAD)

**Supervisor:** Prof. Dr. Peter Schilke

### Summer Intern

Apr 2024 – Jun 2024

*National Institute of Science Education and Research (NISER), Bhubaneswar, India*

**Project:** "Lunar Mineralogy Studies using NASA's LRO Diviner Radiometer"

**Supervisor:** Dr. Guneshwar Thangjam

### Junior Research Fellow

Jun 2023 – Mar 2024

*Indian Institute of Space Science and Technology, Trivandrum, India*

**Project:** "A Global view on Star formation (GLOSTAR) Galactic Plane Survey: 6.7 GHz Methanol Maser Catalog and Variability Study using Effelsberg Data" (Funded by Max-Planck Society)

**Supervisor:** Prof. Dr. Jagadheep D.

### Summer Intern (ASPIRE 2022 Program)

Jun 2022 – Aug 2022

*Anton Pannekoek Institute, University of Amsterdam, The Netherlands (Online)*

**Project:** "A Study of Tides in Hierarchical Triples"

**Supervisor:** Dr. Yan Gao, University of Birmingham, UK

### Summer Intern

Mar 2022 – Jun 2022

*Indian Institute of Astrophysics, Bangalore, India*

**Project:** "A Study of Rising and Setting of Celestial Objects"

**Supervisor:** Dr. Chrispin Karthick

## MSc Lab Compensation Project

Dec 2021 – Feb 2022

*Indian Institute of Astrophysics, Bangalore, India*

**Project:** “Dynamics of an Asteroid Didymos”

**Supervisor:** Dr. Chrispin Karthick

## Summer Intern (Summer Research Fellowship Program, 2020)

Apr 2020 – Jun 2020

*Aryabhata Research Institute of Observational Sciences, Nainital, India*

**Project:** “Astronomical Spectroscopy and Classification of Stars”

**Supervisor:** Dr. Amitesh Omar

## TELESCOPE TIME & OBSERVING EXPERIENCE

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- **Principal Investigator (P.I.), IRAM NOEMA Observatory, Grenoble, France:** Secured 10 hours of observation time (Mar 2026 – Ongoing).
- **Observer, IRAM 30m Telescope:** 50 hours of on-site observation experience for collecting NASCENT-stars complimentary short-spacing data.

## SKILLS

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- **Programming:** Python.
- **Spectral Line Analysis & Modeling:** XCLASS for identification, modeling, and fitting of COMs observational spectra; WEEDS for modeling COMs spectra; GILDAS\CLASS for spectral line analysis.
- **Astrochemical Modeling:** UCLCHEM for modeling chemical evolution of a static cloud (Basic).
- **Data Calibration & Imaging:** GILDAS\CLIC for calibrating NOEMA data; CLASS for reducing IRAM 30m data; GILDAS\MAPPING & IMAGER for imaging NOEMA data.
- **Simulation for Telescopic Observations:** IMAGER for NOEMA in A configuration; CASA for ALMA in 5-10 configuration; ASTRO for planning and preparing observational setup for NOEMA.
- **Software:** XCLASS, CARTA, CASA, GILDAS, DS9.
- **Wavelengths Studied:** Millimeter, Radio, NIR, Optical.

## PUBLICATIONS

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1. **NOEMA Astrochemistry of Cygnus-X Protostars (NASCENT-Stars) I. Continuum analysis of CygX-N53.** D. Panda, T. Csengeri, et al. (To be submitted)
2. **ALMA-IMF. XXII. Role of Core Subfragmentation in the IMF Origin: Hierarchical Mass Cascade in W43-MM1.** F. Motte, N. Le Nestour, R. Veyry, N. Brouillet, T. Nony, B. Thomasson, F. Louvet, I. Joncour, E. Moraux, A. Men'shchikov, T. Yoo, A. Ginsburg, A. Gusdorf, A. M. Stutz, R. Galvan-Madrid, T. Csengeri, R. H. Alvarez-Gutierrez, M. Armante, Y. Bernard, M. Bonfand, S. Chevalier, N. Cunningham, P. Dell'Ova, M. Gonzalez, A. Koley, F. A. Olguin, **D. Panda**, Y. Pouteau, J. Salinas, P. Sanhueza, N. A. Sandoval-Garrido, M. Vaicille-Manet, 2026, *Astronomy & Astrophysics*.\*\* DOI: 10.1051/0004-6361/202660192
3. **Tertiary Tides with Eccentric Orbits.** Y. Gao, T. Boekholt, **D. Panda**, T. Akiba, S. Toonen, 2025, MNRAS. DOI: 10.1093/mnras/staf1460

## WORKSHOPS & SCHOOLS

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**Cosmic Origins Early Career Workshop by NASA**

Oct 21–23, 2025

(Online)

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|----------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| <b>COST NanoSpace – AI in Astrochemistry Training School 2025</b><br><i>Aalto University, Espoo, Finland</i>                           | Aug 4–8, 2025       |
| <b>International Summer School on the ISM of Nearby Galaxies</b><br><i>Banyuls-Sur-mer, France</i>                                     | Jul 21–31, 2025     |
| <b>Galactic Astrochemistry Lecture Series</b><br><i>(Online)</i>                                                                       | Nov 2024 – Feb 2025 |
| <b>12th IRAM Millimeter Interferometry School</b><br><i>IRAM Headquarters, Grenoble, France</i>                                        | Nov 18–23, 2024     |
| <b>Physics and Chemistry of the Interstellar Medium (PCMI)</b><br><i>University of Bordeaux, France</i>                                | Oct 28–30, 2024     |
| <b>(Sub)Millimeter Astronomy: Introduction to ALMA and Indian perspectives</b><br><i>Indian Institute of Science, Bangalore, India</i> | Feb 31, 2024        |
| <b>Short Course on Exoplanets (Pune Knowledge Cluster)</b><br><i>India</i>                                                             | Feb 1–28, 2023      |
| <b>RAD@HOME Citizen Science Program</b><br><i>Indian Institute of Science Education and Research, Berhampur</i>                        | Oct 2019            |
| <b>Physics Training and Talent Search (PTTS)</b><br><i>National Institute of Technology, Mangalore, Karnataka, India</i>               | Jun 1–21, 2018      |

## GRANTS & ACHIEVEMENTS

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1. **DAAD Research Grants in Germany (€3,100)** for a research visit to the University of Cologne, Germany (2026).
2. **PhD Fellowship** from the **Infinity<sup>2</sup> International Graduate** Program, University of Bordeaux (2024-2027).
3. Pondicherry University, India, **Merit Scholarship** & Entrance Test, **All India Rank 10** (2020).
4. Qualified Joint Admission Test for Master's (**IIT-JAM**) (2019).
5. **Higher Education INSPIRE Scholarship (INR 3,60,000)**, Department of Science and Technology (**DST**), India (2017-22).

## MENTORING EXPERIENCE

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| <b>Lily-May Duteil Becker</b><br><i>Bachelor's Student, University of Bordeaux, France</i>                                  | May 1 – Jun 12, 2026 |
| <b>Bachelor's Thesis:</b> "Revealing Embedded Protostars in the Cygnus-X N30 Massive Dense Core at 3mm"                     |                      |
| <b>Saptarshi Das</b><br><i>Master's Student, Indian Institute of Space Science and Technology (IIST), Trivandrum, India</i> | Jun – Jul 2026       |
| <b>Project:</b> "Revealing Embedded Protostars in the Cygnus-X N30 Massive Dense Core at 1mm"                               |                      |

## PEER REVIEW

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| <b>Reviewer, ALMA Proposal Peer-Review System</b><br>Details: Evaluated and reviewed 10 ALMA proposals. | May 2026 |
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## RELEVANT PRESENTATIONS

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1. **Invited Talk:** “NASCENT-stars: Exploring the Molecular Inventory of Massive Protostellar Envelopes in Cygnus-X”, Weekly Group Meeting of Prof. Arshia Jacob, University of Cologne, Germany, May 2026.
2. **Contributory Talk:** “NASCENT-stars: A Study of Fragmentation and Molecular Complexity in Cygnus-X Massive Dense Cores”, Cologne-Bonn Symposium, University of Cologne, Germany, August 2026.
3. **Contributory Talk:** “NASCENT-stars: A Study of Fragmentation and Molecular Complexity in Cygnus-X Massive Dense Cores”, CMF2IMF conference, ESO Headquarters, Garching, Germany, June 2026.
4. **Contributory Talk:** “Origin of Complex Organic Molecules in Massive Star-Forming Regions”, ALMA-IMF Workshop, Online, June 2026.
5. **Institute Talk:** “NASCENT-stars: A Study of Fragmentation and Molecular Complexity in Cygnus-X Massive Dense Cores”, PhD Day, LAB, Bordeaux, France, May 2026.
6. **Contributory Talk:** “NASCENT-stars: Exploring the Molecular Inventory of Massive Protostellar Envelopes in Cygnus-X”, ALMA-IMF Workshop, Online, June 2025.
7. **Invited ECR Talk:** “My Academic Journey to a PhD in Astrophysics”, Govt. Science College, Odisha, India, August 2024.
8. **Poster:** “NASCENT-stars: Exploring the Molecular Inventory of Massive Protostellar Envelopes in Cygnus-X”, COST NanoSpace Training School, Espoo, Finland, August 2025.
9. **Poster:** “A Global View on Star Formation (GLOSTAR) Galactic Plane Survey: 6.7 GHz Methanol Maser Catalogue and Variability Study”, 42nd Annual Meeting of ASI, Bangalore, India, Feb 2024 (Bibcode: 2024asi.confP.167).

## LEADERSHIP & OUTREACH PROGRAMS

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1. Conducted a solar observation session with a 3” reflector and a pinhole projector at the Library of the University of Bordeaux, France (2024).
2. Led a night sky observation event with an 8” reflector telescope at ACSTI, Trivandrum, as part of Rashtriya Avishkar Abhiyan by the Ministry of Education, Govt. of India (2024).
3. Coordinated a public observation session with a refractor telescope to witness the Solar Eclipse at Berhampur University, India (2020).
4. Demonstrated optics experiments and conducted a solar observation session with a refractor telescope on National Science Day at Berhampur University, India (2020).

Last updated on 30<sup>th</sup> Aug 2026